Camera-Less In-Seat Posture Classification & Template Based Pose Reconstruction Using Pressure Sensor Fusion

Siddhant Deore
IIIT Bangalore
Bangalore, India
Siddhant.Deore@iiitb.ac.in

Hrushikesh Sawant
IIIT Bangalore
Bangalore, India
Hrushikesh.Sawant@iiitb.ac.in

Madhav Rao
IIIT Bangalore
Bangalore, India
mr@iiitb.ac.in

***Abstract*—Effective in-cabin monitoring is crucial for enhancing driver and passenger safety; however, systems that incorporate cameras raise significant privacy concerns. This paper presents a novel, privacy-preserving multimodal system that accurately analyzes occupant status using non-visual sensors. Our system integrates a high-resolution (15×15) Velostat pressure grid on the seat, 3 discrete pressure sensors on the back of the seat, and 5 sensors on the seatbelt. We propose a dual-model approach to analyze the pressure data. First, a two-stream Convolutional Neural Network (CNN) classifies occupant posture into three major classes: *good*, *bad*, or *drowsy*, achieving an accuracy of 99.63%. Second, we introduce a novel, camera-less, template-based approach for coarse 2D pose reconstruction via pressure-sensor fusion. A LightGBM model trained on engineered features derived from sensor data reconstructs a 2D stick representation of the occupant across various seating angles with 98% accuracy. By successfully demonstrating high-fidelity posture classification and camera-free, template-based pose reconstruction, our work provides robust and viable information on passengers' sitting postures during commuting. This approach not only ensures occupant safety and comfort but also facilitates real-time visualization of passengers' poses, enabling necessary corrections or adjustments to support long-term health benefits. All model files and the dataset are made freely available for further use by researchers and the engineering community.**

***Index Terms*—Intelligent Vehicles, In-Cabin Monitoring, Pressure Sensors, Pose Reconstruction, Camera-Less Sensing, Privacy-Preserving.**

I. Introduction

Occupant monitoring has become a cornerstone of intelligent vehicle safety, enabling the early detection of risky states such as distraction, musculoskeletal strain, drowsiness, and facilitating interventions ranging from driver alerts to adaptive restraint strategies and cooperative automation handovers [1]–[4]. Various driver monitoring systems comprise camera-based pipelines that incorporate RGB, NIR, and thermal sensors, paired with head/eye pose estimation, blink rate, and facial action units, to infer driver alertness and behavior [5]–[7]. While these approaches have achieved strong accuracy in controlled settings, they raise persistent privacy concerns, struggle in low-light conditions, occlusions, glare, night driving, and camera placement constraints, leading to degraded performance in many real-world driving scenarios and face challenges related to fairness and consent in shared or commercial fleets [8]–[12]. These limitations underscore the need for non-visual sensing methods that safeguard privacy, remain reliable under varying lighting and occlusion conditions, are unaffected by demographic differences, and can be seamlessly integrated for continuous use in real-world vehicles [8], [9], [13].

Non-visual in-cabin sensing has advanced along three complementary directions. First, contact pressure sensing embedded in seats/seat-belts provides a direct, lighting-independent proxy for body posture and load distribution; prior work has shown that pressure maps and even sparse arrays can discriminate posture classes and sitting behaviors with low cost and high reliability [14]–[17]. Second, radio-frequency (RF) sensing, particularly automotive mmWave radar (24–77–60 GHz), enables unobtrusive estimation of cardio-pulmonary activity and micro-motions through cloth and non-metallic materials, supporting occupancy detection, seat assignment, and vitals monitoring under strong privacy guarantees [18]–[21]. Third, simple active infrared elements positioned near the headrest can robustly detect head tilts and off-axis motions with minimal computational requirements, without compromising privacy, and provide a low-latency channel for detecting indicators of driver distraction. This complements pressure and radar streams [22], [23]. Taken together, these modalities form a privacy-preserving technique for the next-generation in-cabin monitoring [13], [18], [24].

Despite these advances, two important gaps remain. First, most studies focus on broad occupant states (e.g., alert vs. drowsy), but understanding fine-grained posture structure and posture variation is critical for long-term health assessment, ergonomics, airbag safety, than a simple yes/no label can provide [14]–[16]. Second, when posture geometry is reconstructed, it is typically vision-led (2D/3D keypoints from cameras), thereby limiting deployability and privacy [5], [7], [10], [25]. These gaps motivate camera-less pose reconstruction, which infers 2D human pose directly from pressure distributions and sparse seatbelt sensors.

In this work, we present a privacy-preserving, multi-modal in-cabin monitoring system that combines a 15×15 Velostat seat grid, five discrete pressure sensors on the seat-belt, and three pressure sensors on the backrest of the seat. We introduce a dual-model architecture: (i) a two-stream CNN that fuses the high-resolution pressure grid and the 8-channel pressure sensor signals on the seat-belt to classify posture to one of the three states - *good*, *bad*, and *drowsy*; and (ii) a camera-less 2D pose reconstruction pipeline that learns to reconstruct a stick representation for four ergonomically relevant seating angles (85°, 95°, 110°, 130°) using only the eight discrete sensors including 5 in seat-belt and 3 in back-rest. While some vehicles provide seatback encoder signals, these measure the mechanical seat angle rather than the occupant's torso orientation; our pressure-based ML mapping therefore targets person-centric posture inference without requiring cameras or OEM seat signals. Ground-truth pose supervision utilizes Google's MediaPipe [26] during dataset creation, but deployment is camera-free. On our dataset, the classifier achieves 99.8% accuracy, and the reconstruction model achieves 98%, demonstrating that fine-grained posture geometry can be recovered without cameras when coupled with targeted feature learning and appropriate temporal smoothing.

Our work consists of three major parts. First, camera-less, high-fidelity posture understanding. We show that spatial pressure metric patterns on the seat backrest and seat belt are sufficient for reliable template-based pose reconstruction across multiple seat angles, addressing a gap left by prior studies that emphasized classification or relied on vision sensors [14], [15], [17], [25]. Second, two-stream pressure fusion for posture safety is developed. A compact CNN integrates pressure grids (2D) on the seat and pressure parameters on the seatbelt (1D) signals with data augmentation (local shifts, noise) to enhance generalization across occupants and seating conditions. Finally, a practical multimodal pipeline for Advanced Driver Assistance Systems (ADAS). We pair mmWave radar for non-contact vital signs and IR head-tilt sensing with pressure-based posture analysis, forming a holistic, privacy-preserving monitoring suite for non-visual-based occupant state estimation. This work specifically addresses the challenges of shared mobility, regulatory requirements, and user acceptance. By minimizing visual capture, it reduces privacy risks, eases data governance requirements, and lowers computational overhead, while maintaining observability under occlusion and at night [8], [9], [13], [24].

II. Proposed Design

To address the limitations of camera-based monitoring, we have engineered and implemented a multimodal, non-visual sensor suite for improved in-cabin analysis. Our approach is focused on capturing high-resolution pressure distribution data, which helps in indicating and detecting the passenger's posture. The entire system is designed to be easily integrated into the vehicle interior, ensuring that the passenger's comfort and privacy are maintained at all times.

A. Hardware Design

The core of our system is a custom-designed sensor apparatus integrated into a standard vehicle seat. The complete hardware configuration, as illustrated in Fig. 1, comprises three primary sensor types selected for their specific capabilities and discrete nature.

Fig. 1. Placement of the 15×15 pressure sensor grid, 5 seatbelt and 3 seat-back pressure sensors within the designed vehicle seat.

*1) **Pressure Sensing System**:* The primary data source is a network of piezo-resistive Velostat sensors, selected for their large range and durability. This network is bifurcated to capture micro postural details:

- **High-Resolution Seat Grid:** A 15×15 grid of Velostat sensors, creating 225 individual pressure points, is positioned on the top of the seat cushion. This dense array provides a detailed pressure map that captures the distribution of the passenger's weight. This is crucial for distinguishing subtle postural shifts that indicate discomfort or the onset of drowsiness.
- **Discrete Body and Belt Sensors:** Eight additional Velostat sensors are strategically placed on key contact areas. Three are located on the seat-back to measure torso pressure and spinal alignment, while the other five are integrated into the seat belt to measure tension and body position relative to the restraint system. This provides explicit and crucial information about leaning, slouching, and bracing behaviors.

*2) **Microcontroller Unit**:* A central microcontroller unit (MCU) is responsible for data aggregation. It samples data from all 233 pressure points at a synchronized rate of 1 Hz (115200 baud for HR sensor). The MCU then packages this information into a unified data frame for transmission to a processing unit, ensuring synchronized sampling across all sensing modalities for real-time analysis.

B. Dataset Collection

A diverse dataset was collected to train and validate machine learning (ML) models. The data collection protocol was designed to capture a wide range of realistic driving postures

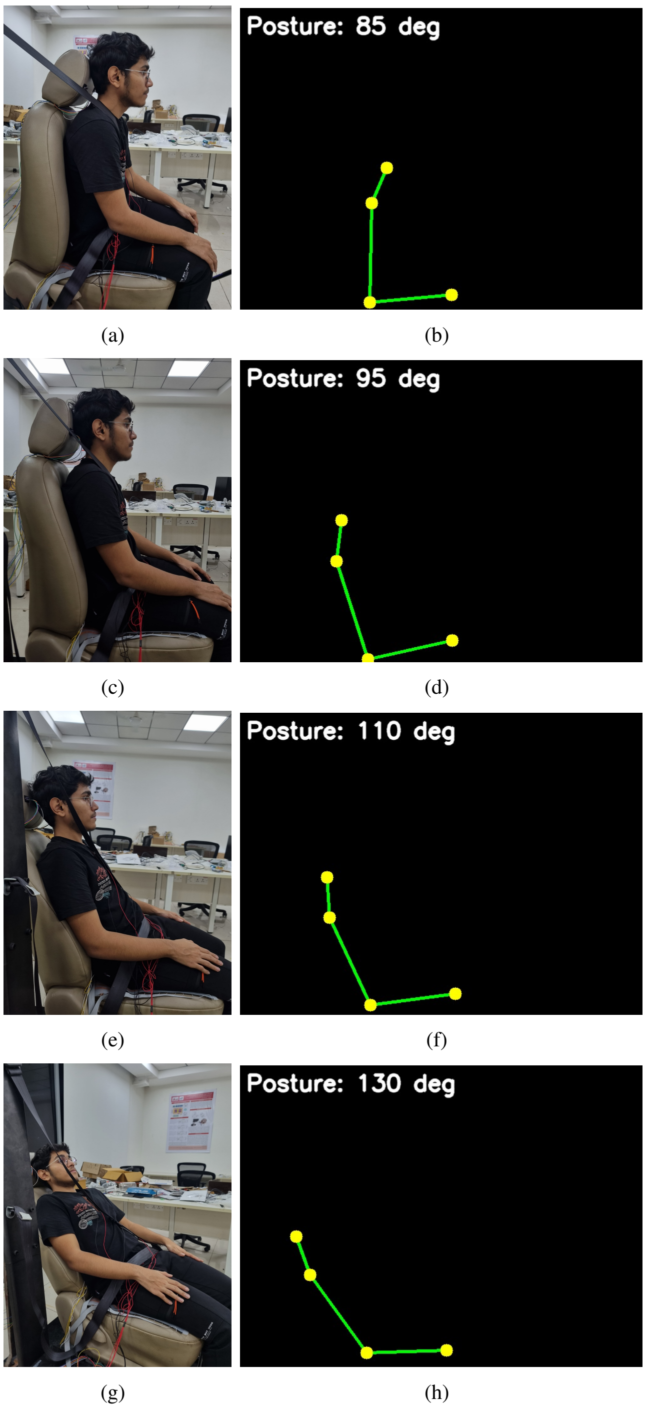


Fig. 2. Reconstructed Stick representation corresponding to different Human Poses - a) 85 °, b) 95°, c) 110°, d) 130°.

and behaviors. A total of 10 participants (aged between 18-30 years and spanning a varying range of heights and weights to ensure model generalizability) were recruited for the study. Ethical consent was obtained from participants for the use of their posture data to validate the proposed design concept. Approval from the institute's research board was also obtained to conduct these experiments. The data was collected in accordance with the Helsinki Declaration of 1975, as revised in 2000 [27]. For the posture classification model, each participant was instructed to hold three distinct, labeled postures for 2 minutes each, as detailed below.

- **Good Posture:** An upright, alert position with clear back support was considered the best posture. A similar posture is advised for the person in the driving position with hands on the steering wheel [28].
- **Bad Posture:** A set of common non-ideal postures, including slouching, leaning heavily on the center console, and sitting twisted [28].
- **Drowsy Posture:** Postures indicative of fatigue, such as a tilted head against the window, a dropped chin, and a slumped upper body [28].

Participants were not constrained to rigidly defined poses and were permitted to adopt natural postures within each category based on their personal comfort and perception.

For the posture reconstruction model, participants were instructed to sit in four postures corresponding to distinct seat recline angles at the hip joint, namely 85°, 95°, 110°, and 130°. This ensures the model learns to reconstruct poses across a full range of typical seat adjustments. A critical part of our methodology was the generation of ground-truth data for this reconstruction task. Because our final system is cameraless, we used an external high-frame-rate camera solely for data collection during this phase. Video footage of the participants was processed using Google's MediaPipe Pose framework [26] to extract the 2D pixel coordinates of 4 body keypoints for each frame. The timestamped 2D pose data was then precisely synchronized with the corresponding multi-channel pressure sensor data. This synchronization process created a high-quality, labeled dataset in which each frame of 8-channel sensor readings was paired with its ground-truth 2D human pose, enabling supervised training of our novel reconstruction model.

III. Model Architecture

To effectively process the data from multiple sensors, we developed two distinct machine learning models, each adapted for its specific task: one for high-accuracy posture classification and another for efficient, real-time template-based pose reconstruction.

A. *Posture Classification Model*

For the task of classifying passengers' posture into good, bad, or drowsy states, we designed a Two-Stream Convolutional Neural Network (CNN), implemented in PyTorch. This architecture is well-suited to handling the heterogeneity of our pressure data.

*1) **Input Streams**:* The model processes the two types of pressure data in parallel:

- **1D Belt/Body Stream:** The data from the eight discrete sensors is treated as an 8-channel, 1D signal. This stream consists of two 1D convolutional layers with BatchNorm and ReLU activation, designed to learn spatial patterns along the occupant's torso and seatbelt.
- **2D Grid Stream:** The 225 readings from the seat grid are reshaped into a 15×15 single-channel image. This stream employs a series of 2D convolutional layers and Max Pooling to learn spatial features from the pressure distribution map, in a manner similar to conventional image processing.

Each input window is a two-stream pair with shapes $\mathbf{b} \in \mathbb{R}^{8\times 1}$ (belt and seat/back) and $\mathbf{g} \in \mathbb{R}^{1\times 15\times 15}$ (grid), batched as $(N, 8, 1)$ and $(N, 1, 15, 15)$ respectively; $\mathbf{b}$ contains eight simultaneous sensor readings at one timestep and $\mathbf{g}$ is the single-channel 15×15 pressure map.

*2) **Feature Fusion and Classification**:* The feature vectors extracted from both streams are concatenated. This combined vector is then passed through a final classifier head comprising fully connected layers with Dropout for regularization, which outputs the final classification probabilities for the three posture states. The classifier model architecture is shown in Fig. 3.

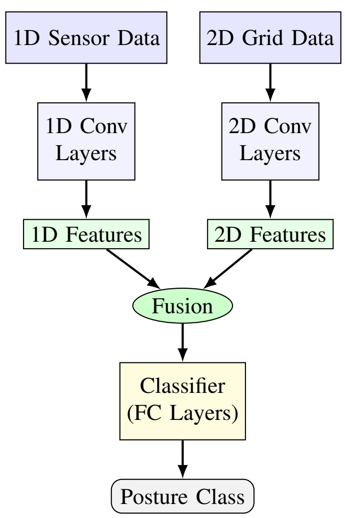


Fig. 3. High-level architecture of the Two-Stream CNN for posture classification.

B. Pose Reconstruction Model

For the task of camera-less 2D pose reconstruction, we prioritized computational efficiency and real-time performance. Thus, this model uses only data from the three sensors mounted on the seat back and the five sensors mounted along the seat belt. Each input sample to the model consists of eight pressure readings from the three seat-back and five seat-belt sensors, corresponding to one time step.

*1) **Intelligent Feature Engineering**:* Instead of using raw sensor values, we first apply feature engineering to create a more descriptive representation of the data. For each time step,

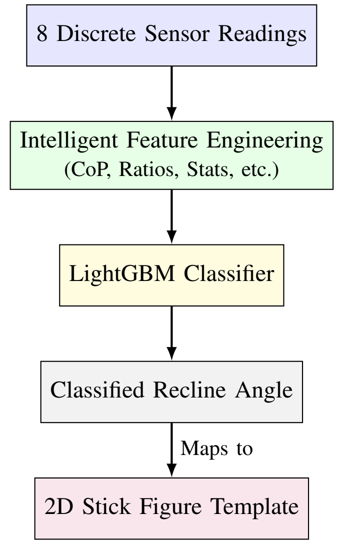


Fig. 4. Flowchart of the LightGBM-based model for 2D Pose reconstruction.

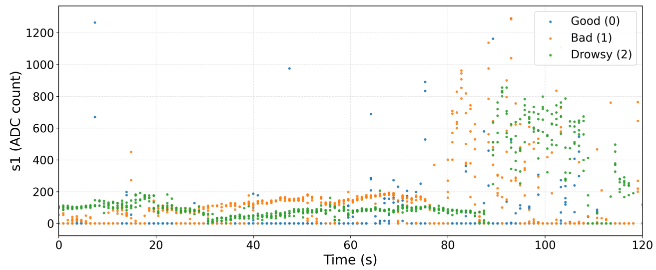
TABLE I
SUMMARY OF THE 21 ENGINEERED FEATURES FOR THE LIGHTGBM MODEL.

Category	Feature Name	Formula	Count
Base	seatbelt_i ($i \in \{0 \ldots 7\}$)	S_i	8
Statistical	mean	$\mu = \frac{1}{8}\sum_{i=0}^{7} S_i$	1
Statistical	std	$\sigma = \sqrt{\frac{1}{8}\sum_{i=0}^{7}(S_i - \mu)^2}$	1
Statistical	max	$\max(S_0, \ldots, S_7)$	1
Statistical	sum	$P_{total} = \sum_{i=0}^{7} S_i$	1
CoP	cop_1d	$\frac{\sum_{i=0}^{7}(S_i \times i)}{P_{total}}$	1
Ratio	top_bottom_ratio	$\frac{\sum_{i=0}^{3} S_i + \epsilon}{\sum_{i=4}^{7} S_i + \epsilon}$	1
Gradient	diff_i_i+1 ($i \in \{0 \ldots 6\}$)	$S_{i+1} - S_i$	7
		Total Features	**21**

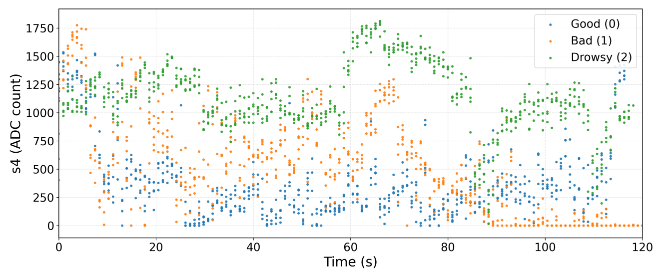
we calculate a set of 21 features from the eight sensor inputs, including:

- **Statistical Features:** Mean, standard deviation, maximum, and sum of the sensor readings.
- **Center of Pressure (CoP):** A 1D CoP is computed to determine the vertical center of pressure along the seat-back.
- **Ratio Features:** Ratios between the upper and lower sensor groups.
- **Gradient Features:** Differences between adjacent sensor readings to capture pressure gradients.

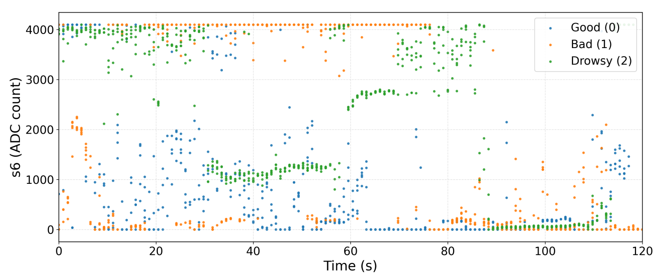
*2) **LightGBM Classifier**:* A Light Gradient Boosting Machine (LightGBM) model is trained on these engineered features. LightGBM is a highly efficient, tree-based gradient-boosting framework known for its speed and low memory usage. The model is trained to classify the current sensor readings into one of the four predefined recline angles (85°, 95°, 110°, 130°). The output is not a direct regression of keypoints but a classification of the overall pose, which then maps to a predefined set of coordinates that, in turn, yields the 2D pose reconstruction in the form of a stick figure. The overall model architecture is depicted in Fig. 4.



(a)



(b)



(c)

Fig. 5. Time-series variation of three pressure sensors - two from the seat-belt and one from the back of the seat for different posture classes: (a) Sensor s1, (b) Sensor s4, and (c) Sensor s6.

IV. Experiments and Results

The performance of both models was rigorously evaluated on the collected dataset. The dataset was randomly shuffled and split into training (80%) and test (20%) sets, with stratification to ensure proportional representation of each class in both sets.

A. Exploratory Data Analysis

Before training the models, we analyzed the raw pressure data to understand relations between the classes. Fig. 5 shows the time-series variation of the three seat-belt and back sensors for each posture class. Although the mean pressure levels differ slightly, there is overlap among the readings, indicating that a simple threshold-based classification would be insufficient to capture the data's complexity. Therefore, a multidimensional approach, such as the proposed CNN architecture, was necessary to capture these complex relationships among sensors.

TABLE II
Characteristics Report for Posture Classification Model.

Class	Precision	Recall	F1-Score
Good	99.77	99.55	99.66
Bad	99.33	99.56	99.45
Drowsy	99.78	99.78	99.78
Accuracy		**99.63**	

TABLE III
Characteristics Report for Reconstruction Model.

Class	Precision	Recall	F1-Score
130 degrees	96.97	96.97	96.97
110 degrees	98.47	98.47	98.47
95 degrees	97.73	97.73	97.73
85 degrees	100	100	100
Accuracy		**98.29**	

B. Posture Classification Performance

The Two-Stream CNN was trained for 50 epochs using the Adam optimizer and a cross-entropy loss function. The model achieved an overall classification accuracy of **99.63%** on the unseen test data. A detailed breakdown of the performance for each class is presented in Table II. The precision, recall, and F1-score for all three classes are high, indicating that the model is both precise and robust. This result validates the model's ability to effectively learn discriminative features from the fused 1D and 2D pressure data for fine-grained posture detection.

C. Pose Reconstruction Performance

The LightGBM model for pose reconstruction was evaluated for its ability to accurately classify the occupant's recline angle from 8-channel sensor data. The model achieved an overall accuracy of **98.29%** on the test set, also presented in the Table III. The high accuracy demonstrates that the engineered features provide sufficient information to robustly distinguish between different seating angles.

In the real-time implementation, the model's output is stabilized using a Holt's double exponential smoothing filter to prevent flickering between states. The classified angle is then used to trigger a smooth, animated transition to the corresponding 2D stick-representation template, providing a fluid and intuitive visual representation of the occupant's pose.

V. Conclusion

This paper introduces a novel multimodal, privacy-preserving system for intelligent in-cabin monitoring. By using a high-resolution pressure-sensing suite, our system successfully achieves two key objectives: (1) highly accurate, fine-grained classification of occupant posture, and (2) a real-time 2D template-based human pose reconstruction without the use of any cameras. Our results demonstrate that pressure-based sensing is a powerful alternative to optical systems, providing rich data for enhancing vehicle safety and comfort. The cameraless approach presented here offers a clear path forward for developing advanced driver and passenger monitoring systems that are not only effective but also universally acceptable. Future work could employ regression-based models to predict the exact 2D coordinates of key body points, enabling the reconstruction of a continuous range of postures. Additionally, it will extend the framework to dynamic, real-time driving scenarios using continuous data streams and temporal models to capture posture transitions and transient states. Robust operation in such settings will require motion-aware processing, including temporal filtering, frequency-based noise suppression, and sensor fusion strategies to mitigate vehicle-induced disturbances. All model files and datasets are made freely available at [29] for further use by researchers and the engineering community.

REFERENCES

[1] M. Roth and D. M. Gavrila, "intrapose: Monocular driver 6 dof head pose estimation leveraging camera intrinsics," *IEEE Transactions on Intelligent Vehicles*, vol. 8, no. 8, pp. 4057–4068, 2023.

[2] ——, "Dd-pose - a large-scale driver head pose benchmark," in *2019 IEEE Intelligent Vehicles Symposium (IV)*, 2019, pp. 927–934.

[3] S. Abtahi, B. Hariri, and S. Shirmohammadi, "Driver drowsiness monitoring based on yawning detection," in *2011 IEEE International Instrumentation and Measurement Technology Conference*, 2011, pp. 1–4.

[4] A. Koesdwiady, R. Soua, F. Karray, and M. S. Kamel, "Recent trends in driver safety monitoring systems: State of the art and challenges," *IEEE Transactions on Vehicular Technology*, vol. 66, no. 6, pp. 4550–4563, 2017.

[5] Y. Chen and Z. Li, "Flextrack3d: Advanced single-camera 3d human pose tracking with flexposenet and zoedepth," *IEEE Access*, vol. 12, pp. 171 334–171 347, 2024.

[6] W. Deng and R. Wu, "Real-time driver-drowsiness detection system using facial features," *IEEE Access*, vol. 7, pp. 118 727–118 738, 2019.

[7] S. Jha, N. Al-Dhahir, and C. Busso, "Driver visual attention estimation using head pose and eye appearance information," *IEEE Open Journal of Intelligent Transportation Systems*, vol. 4, pp. 216–231, 2023.

[8] F. Cunico, S. Aldegheri, A. Avogaro, M. Boldo, N. Bombieri, L. Capogrosso, A. Caputo, D. Carra, S. Centomo, D. S. Cheng, E. Cinquetti, M. Cristani, M. D. Marchi, F. Demrozi, M. Emporio, F. Fummi, L. Geretti, S. Germiniani, A. Giachetti, F. Girella, E. Martini, G. Menegaz, N. Muijs, F. Paci, M. Panato, G. Pravadelli, E. Quintarelli, I. Siviero, S. F. Storti, C. Tadiello, C. Turetta, T. Villa, N. Zannone, and D. Quaglia, "Enhancing safety and privacy in industry 4.0: The ice laboratory case study," *IEEE Access*, vol. 12, pp. 154 570–154 599, 2024.

[9] F. Znidi, M. Morsy, and H. Rathore, "Future on wheels: Safeguarding privacy in tomorrow's connected vehicles-future-sp," *IEEE Access*, vol. 12, pp. 179 857–179 878, 2024.

[10] M. Zhao, T. Li, M. Abu Alsheikh, Y. Tian, H. Zhao, A. Torralba, and D. Katabi, "Through-wall human pose estimation using radio signals," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018.

[11] P. Kedia, P. Aggarwal, and M. Rao, "Design and development of an autonomous in-seat passenger state identification in a modern vigilance enabled public transportation system," in *2019 IEEE International Conference on Vehicular Electronics and Safety (ICVES)*, 2019, pp. 1–6.

[12] P. Nandi, A. Mishra, P. Kedia, and M. Rao, "Design of a real-time autonomous in-cabin sensory system to detect passenger anomaly," in *2020 IEEE Intelligent Vehicles Symposium (IV)*, 2020, pp. 202–206.

[13] J. P. Van Marter, A. G. Dabak, A. V. Mani, S. Rao, and M. Torlak, "A deep learning approach for in-vehicle multi-occupant detection and classification using mmwave radar," *IEEE Sensors Journal*, vol. 24, no. 20, pp. 33 736–33 750, 2024.

[14] C. Ma, W. Li, R. Gravina, and G. Fortino, "Posture detection based on smart cushion for wheelchair users," *Sensors*, vol. 17, no. 4, 2017. [Online]. Available: https://www.mdpi.com/1424-8220/17/4/719

[15] L. Yuan and J. Li, "Smart cushion based on pressure sensor array for human sitting posture recognition," in *2021 IEEE Sensors*, 2021, pp. 1–4.

[16] Q. Hu, X. Tang, and W. Tang, "A smart chair sitting posture recognition system using flex sensors and fpga implemented artificial neural network," *IEEE Sensors Journal*, vol. 20, no. 14, pp. 8007–8016, 2020.

[17] S. Suzuki, M. Kudo, and A. Nakamura, "Sitting posture diagnosis using a pressure sensor mat," in *2016 IEEE International Conference on Identity, Security and Behavior Analysis (ISBA)*, 2016, pp. 1–6.

[18] Z. Yang and Z. Bao, "Short-time heart rate estimation based on 60-ghz fmcw radar," in *2023 IEEE MTT-S International Wireless Symposium (IWS)*, 2023, pp. 1–3.

[19] C. Li, V. M. Lubecke, O. Boric-Lubecke, and J. Lin, "A review on recent advances in doppler radar sensors for noncontact healthcare monitoring," *IEEE Transactions on Microwave Theory and Techniques*, vol. 61, no. 5, pp. 2046–2060, 2013.

[20] D. Bresnahan and Y. Li, "Driver head motion monitoring using a mm-wave fmcw radar," in *2021 IEEE Texas Symposium on Wireless and Microwave Circuits and Systems (WMCS)*, 2021, pp. 1–4.

[21] C. Schwarz, H. Zainab, S. Dasgupta, and J. Kahl, "Heartbeat measurement with millimeter wave radar in the driving environment," in *2021 IEEE Radar Conference (RadarConf21)*, 2021, pp. 1–6.

[22] E. Murphy-Chutorian, A. Doshi, and M. M. Trivedi, "Head pose estimation for driver assistance systems: A robust algorithm and experimental evaluation," in *2007 IEEE Intelligent Transportation Systems Conference*, 2007, pp. 709–714.

[23] A. Firintepe, M. Selim, A. Pagani, and D. Stricker, "The more, the merrier? a study on in-car ir-based head pose estimation," in *2020 IEEE Intelligent Vehicles Symposium (IV)*, 2020, pp. 1060–1065.

[24] H. Kong, C. Huang, J. Yu, and X. Shen, "A survey of mmwave radar-based sensing in autonomous vehicles, smart homes and industry," *IEEE Communications Surveys Tutorials*, vol. 27, no. 1, pp. 463–508, 2025.

[25] D. Kang and D. Kang, "Head pose-aware regression for pupil localization from a-pillar cameras," *IEEE Access*, vol. 12, pp. 11 083–11 094, 2024.

[26] C. Lugaresi, J. Tang, H. Averbuch-Erez, J. Taylor, C. Smith, J. F. Higgins, M. Grundmann, J. Wei, D. Keysers, W. Macherey, M. S. Ryoo, G. Papandreou, H. Adam, and V. Rajendran, "Mediapipe: A framework for building perception pipelines," *arXiv preprint arXiv:1906.08172*, 2019.

[27] R. E. Ashcroft, "The declaration of helsinki," in *The Oxford Textbook of Clinical Research Ethics*. Oxford University Press, 04 2008. [Online]. Available: https://doi.org/10.1093/oso/9780195168655.003.0014

[28] M. Nadeem, E. Elbasi, A. I. Zreikat, and M. Sharsheer, "Sitting posture recognition systems: Comprehensive literature review and analysis," *Applied Sciences*, vol. 14, no. 18, 2024. [Online]. Available: https://www.mdpi.com/2076-3417/14/18/8557

[29] https://github.com/hades534/Camera-Less-In-Seat-Posture-Classification-Pose-Reconstruction-Using-Pressure-Sensor-Fusion.